Environmental Science & Technology

pubs.acs.org/est

Article

Coupling Iron Sludge Addition and Intermittent Aeration for Achieving Simultaneous Methanogenesis, Feammox, and Denitrification in a Single Reactor Treating Fish Sludge

Qing Xia,[⊥] Fei Liu,[⊥] Shengrui Sun, Wenli Huang, Ziwen Zhao, Fei Yang, Zhongfang Lei, Weiwei Huang,* and Xuesong Yi*

Cite This: *Environ. Sci. Technol.* 2023, 57, 15065−15075

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: An integrated anaerobic digestion system for the simultaneous removal of carbon and nitrogen from fish sludge was developed by coupling iron sludge supplementation with intermittent aeration. In terms of nitrogen removal, Fe(III) in iron sludge could trigger Feammox reactions and intermittent aeration could drive the Fe(II)/Fe(III) cycle to sustain continuous ammonia removal. Mass balance analysis suggested that nitrate was the main product of Feammox, which was subsequently removed through heterotrophic denitrification. In terms of carbon removal, the Fe(III)-induced dissimilatory iron reduction (DIR) process significantly promoted fish sludge hydrolysis and provided more simple organics for methanogens and denitrifiers, but aeration showed a negative impact on methanogenesis. To promote nitrogen removal and avoid serious methanogenesis inhibition, different aeration intensities were studied. Results showed that compared with the control without aeration or iron sludge addition, aeration for 5 min every 3 days (150 mL/min) contributed to a 29.0% lower NH_4^+-N concentration and a 12.1% lower total chemical oxygen demand level on day 28, and the decline in methane yield was acceptable (only 13.5% lower). Simultaneous methanogenesis, Feammox, and denitrification in a single reactor treating fish sludge were achieved, which provides a simple and low-cost strategy for the treatment of organic wastewater.



KEYWORDS: *fish sludge, nitrogen removal, carbon removal, CH_4 recovery, Fe(II)/Fe(III) cycle, metagenomic analysis*

1. INTRODUCTION

Huge amounts of fish sludge, consisting of fecal solids and undigested feed, are generated each year from the freshwater aquaculture industry globally and require proper treatment and utilization. Anaerobic digestion (AD) is an environmentally friendly process that converts organic matter into combustible methane (CH_4), and has long been practiced for the treatment of fish sludge.[1] However, the digestates containing high levels of ammonia require further treatment before discharge. The normal practice is to set up biological nitrogen (N) removal units after AD, which not only complicates the treatment procedure but also increases the energy consumption, land use, and construction cost.[2,3] Therefore, it is necessary to develop integrated anaerobic treatment systems for the simultaneous removal of carbon (C) and N.

In recent years, researchers have successfully coupled methanogenesis and denitrification in a single reactor, achieving simultaneous removal of organic C and nitrate/nitrite.[4,5] The small molecular organic substances produced from the fermentation process can provide carbon sources for the denitrifiers, and the alkalinity generated during the denitrification reactions can create a more stable environment for the methanogens.[4,5] Despite the merits of simultaneous methanogenesis and the denitrification process, the organic N rich in fish sludge is mainly decomposed to ammonia during AD, which cannot be directly utilized by denitrifiers. The conventional ammonia oxidation and nitrite oxidation processes generally function at a dissolved oxygen (DO) concentration no less than 0.4 and 3.0 mg/L, respectively,[6] which are difficult to occur in AD reactors. Therefore, finding effective strategies to convert ammonium-nitrogen (NH_4^+-N) into nitrite-nitrogen (NO_2^--N) or nitrate-nitrogen (NO_3^--N) under anaerobic environments is the key to simultaneously

Received: April 25, 2023
Revised: September 14, 2023
Accepted: September 15, 2023
Published: September 29, 2023



Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

ACS Publications

© 2023 American Chemical Society

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

achieving N removal and CH_4 recovery in a single reactor treating fish sludge.

Anaerobic ammonium oxidation coupling with Fe(III) reduction (Feammox) has recently been discovered as a new pathway for N removal.[7−9] During this process, Fe(III) was utilized as the electron acceptor to oxidize NH_4^+-N into nitrogen gas (N_2), NO_2^--N, or NO_3^--N. Theoretically, the formed NO_x^- can be further utilized by heterotrophic denitrifiers in the presence of an adequate amount of organic C.[10] Therefore, the implementation of Feammox in AD reactors is expected to advance the conversion of NH_4^+-N into N_2 by coupling with denitrification. Iron-reducing bacteria (IRBs) are identified as the key mediators of Feammox process, and various Fe(III) compounds including Fe_2O_3, $Fe(OH)_3$, $FeCl_3$, pyrite, magnetite, and goethite have been proven to be effective for enriching IRBs and triggering Feammox reactions.[10,11] Iron sludge that mainly consists of $Fe(OH)_3$ is a type of waste generated from Fenton reactions (for advanced treatment of the aquaculture effluent) or iron-salt-based flocculation (for solid−liquid separation). The good availability, relatively high solubility, large specific surface area, high microbial accessibility, and low crystallinity of iron sludge make it an ideal Fe(III) source. It is noteworthy that the Feammox efficiency is positively correlated with the microbial reducible Fe(III) content.[11] To sustain successive Feammox reactions for NH_4^+-N removal from N-rich fish sludge, a large amount of exogenous Fe(III) is needed, which is impractical from an economical or environmental point of view. What is more, the formed Fe(II) could attach to the surface of Fe(III) compounds, changing their thermodynamic properties and accelerating their transformation into more crystalline minerals,[12] which further decreases the microbial reducible Fe(III). To increase the utilization efficiency of Fe(III), it is important to find effective strategies to convert Fe(II) into Fe(III) in a timely manner and reutilize it in the next round of Feammox.

Oxygen (O_2) is a cheap, readily available, and effective oxidant. The oxidation of Fe(II) into Fe(III) can occur at a very low DO level of less than 0.02 mg/L,[13] which is much lower than that of conventional nitrification process. A low level of DO would not seriously inhibit the AD process. Instead, the dosing of a small amount of air or O_2 (DO range of 0.1−1.0 mg/L) into the AD system could effectively enhance the degradation of organics through promoting the growth and metabolisms of facultative bacteria.[14] To achieve effective N removal from the AD reactor without significantly deteriorating the methanogenesis performance, the aeration intensity needs to be optimized. Several studies have employed Feammox-based reactors for the simultaneous removal of ammonia and organics by creating a DO fluctuation environment. Yang et al. reported that by periodically increasing the DO to 0.4 mg/L through intermittent aeration, Fe(III) was regenerated and involved in the next run of Feammox and dissimilatory iron reduction (DIR).[15] After 90 days of reaction, the overall removal efficiencies of total nitrogen (TN) and total chemical oxygen demand (TCOD) were 98.5 and 43.2%, respectively. Ma et al. observed the Fe(III)/Fe(II) redox cycle in a Feammox-based up-flow anaerobic reactor using limonite as the filter medium.[16] The initial DO was 0.58 mg/L, which gradually decreased as the water flowed upward. Results showed that more than 55.0% of influent TCOD was removed in the lower layer of the reactor, while 60.2% of NH_4^+-N was removed in the middle layer. These previous studies are mainly focused on the removal of N and C; little attention was paid to the CH_4 recovery efficiency or the syntrophic relationship between Feammox and denitrification. To our acknowledgment, the feasibility of coupling methanogenesis, Feammox, and denitrification in a single reactor has not been studied yet. The microbial interactions in the integrated system and the effect of aeration intensity on the conversion of C and N remain to be explored.

In this study, a novel simultaneous methanogenesis, Feammox, and denitrification (SMFD) system was developed for the treatment of fish sludge by coupling iron sludge addition with intermittent aeration. The performances of CH_4 production and N removal at different aeration intensities (5, 10, and 15 min every 3 days at a constant rate of 150 mL/min) were systematically investigated. Microbial analysis was also carried out to investigate the microbial community distributions, diversities, functional traits, and microbial interactions in the SMFD system.

2. MATERIALS AND METHODS

2.1. Materials. The iron sludge used in this study was prepared through a Fenton process in a 1-L glass beaker at room temperature. Specifically, 69.5 g of $FeSO_4{\cdot}7H_2O$ was dissolved in 500 mL of deionized water. After complete dissolution of $FeSO_4{\cdot}7H_2O$, 85 mL of a H_2O_2 solution (30% mass fraction) was added. The molar ratio of H_2O_2 to Fe^{2+} was 3:1.[17] After 2 h of reaction, the system pH was adjusted to 7.0 with NaOH solution (1 mol/L). After precipitation for about 24 h, the liquid was centrifuged at 8000 rpm for 10 min. The collected iron sludge with a moisture content of about 55% was stored at 4 °C before use.

Fresh fish sludge was collected from a tilapia fish farm located in Wenchang (China), transferred to the laboratory, and immediately stored at 4 °C. The seed sludge was taken from a commercial-scale digester in Baishamen municipal wastewater treatment plant (Haikou, China), and was mixed thoroughly with fish sludge and incubated at 35 °C in a lab-scale batch digester (40-L). During the incubation period, 10 g of iron sludge was added to the digester on a weekly basis for the enrichment of IRBs. To guarantee complete decomposition of the labile organics, the incubation process lasted for 2 months. The mature digestate from this digester was used as the inoculum in this work. Biogas production from the inoculum during subsequent experiments was negligible. Basic characteristics of fish sludge and the inoculum are listed in Table S1.

2.2. Batch Experiments. In this study, a series of identical cylindrical glass reactors (188 mm in diameter and 278 mm in height) was used for the batch AD experiments. As shown in Figure S1, two types of AD systems were set up, namely, the aerated system and control system. For the aerated system, air was intermittently bubbled into the liquid via a gas diffuser at the bottom of the reactor. Intermittent aeration was achieved by using an electric timing control switch (Bull, China) and a 12 V DC micro gas pump (Xinweicheng, China). The air flow rate was maintained at a constant level of 150 mL/min. Liquid samples were taken periodically from the aerated reactors for DO analysis during one aeration cycle. The digesters were sealed with silicon stoppers, and the biogas produced from each reactor was collected in a gas sampling bag for analysis every day. Before each aeration cycle, the collected biogas was quantified. After aeration, the air captured by the gas bag was

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

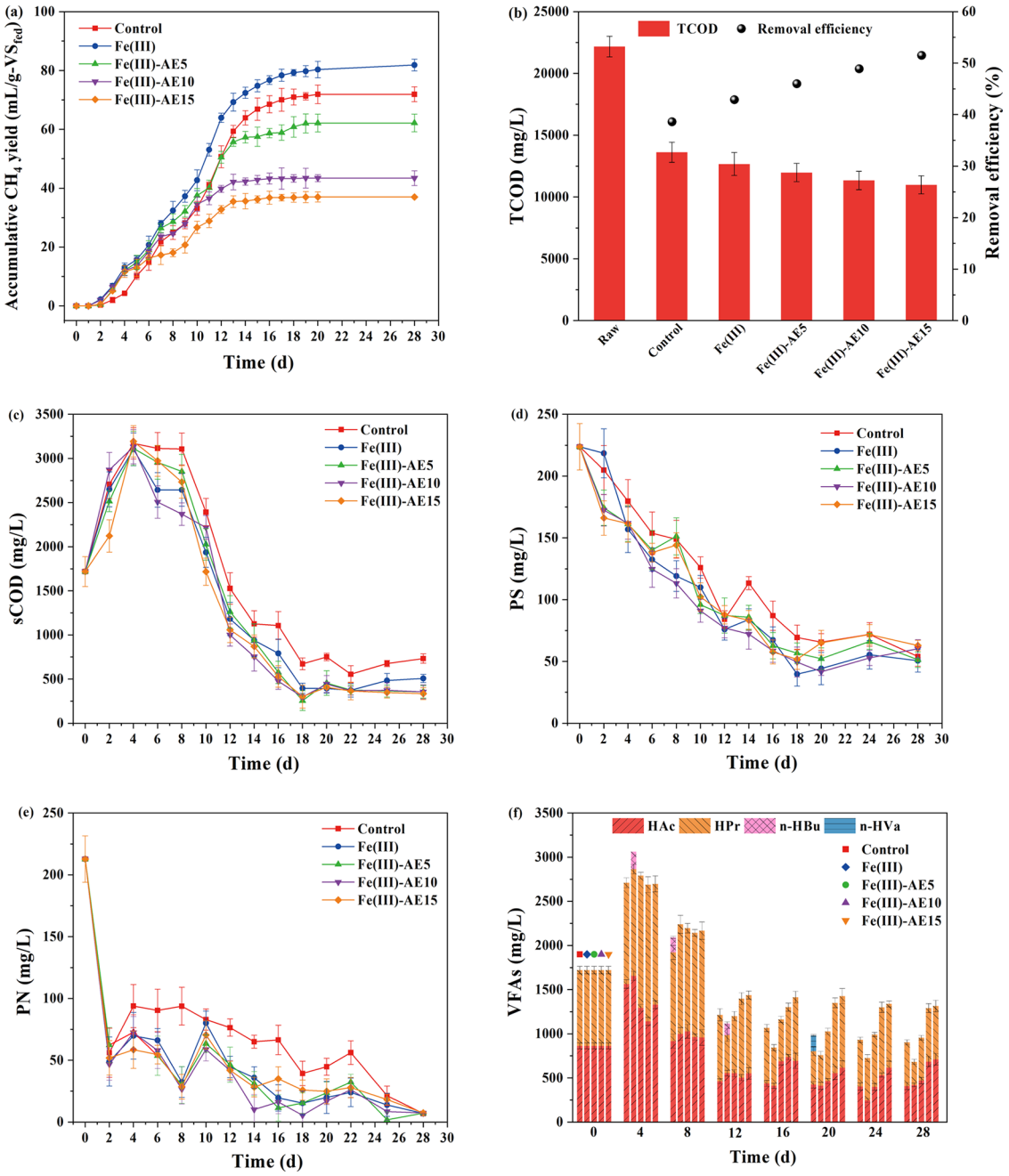


Figure 1. Concentrations of (a) accumulative CH_4 yield, (b) TCOD, (c) sCOD, (d) PS, (e) PN, and (f) VFAs including acetic acid (HAc), propionic acid (HPr), n-butyric acid (n-HBu), and n-valeric acid (n-HVa).

discharged immediately to minimize the dilution effect of air on biogas.

Each reactor received 700 g of fish sludge and 300 g of inoculum (fish sludge to inoculum ratio of 3:1 based on volatile solids (VS)), and it was diluted with deionized water to a final volume of 1.5 L. Main characteristics of the mixture are listed in Table S1. To study the sole effect of iron sludge addition, 2 experimental groups were set up, namely, the Fe(III) group with the addition of 10 g of iron sludge and the control group without iron sludge. To study the effect of different aeration intensities, 3 experimental groups added with 10 g of iron sludge were aerated for 5, 10, and 15 min every 3 days since day 4 at a constant rate of 150 mL/min, and were denoted as Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15, respectively. Each experimental group was prepared in duplicate. All reactors were flushed with N_2 for 20 min prior to the experiment and were incubated at 35 °C for 28 days.

To evaluate the possible contribution of aeration to NH_4^+ conversion/removal, an additional group of AD experiments was performed with aeration (150 mL/min for 10 min every 3 days) and without Fe(III) addition (AE10). Other experimental conditions were the same as the Fe(III)-AE10 group.

2.3. Analysis Methods. The biogas production volume was recorded by using a syringe. The biogas compositions were analyzed using a gas chromatograph (FULI, GC9790Plus) with a thermal conductivity detector (TCD) and a packed column. Argon (purity >99.999%) was used as the carrier gas.

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

The injector, column, and detector temperatures were set at 80, 50, and 80 °C, respectively.

For the determination of the key process parameters, liquid samples were taken periodically. The pH values were measured by using a pH meter (AS600, ASONE, Japan). The DO was measured by using a DO sensor (ST400D, OHAUS, China). The TCOD, total solids (TS), VS, and TN were measured at the beginning and the end of the experiments according to Standard Methods.[18] The organic C and N contents of the inoculum, fish sludge, and mixture were analyzed by an elemental analyzer (PerkinElmer 2004 CHN, USA). For analysis of soluble compounds, liquid samples were centrifuged at 9000 rpm for 20 min and filtered through a 0.22-μm membrane. A gas chromatograph (FULI, GC9790Plus) equipped with a flame ionization detector (FID) and a capillary column (30 m, 0.32 mm I.D., 0.5 μm) was used to determine the concentrations of volatile fatty acids (VFAs). The column temperature was increased from 80 to 180 °C at a rate of 10 °C/min, and the temperatures of the injector and detector were 250 and 280 °C, respectively. Argon gas (purity >99.999%) was applied as the carrier gas. The filtrate was analyzed for soluble polysaccharides (PS), proteins (PN), and soluble chemical oxygen demand (sCOD) as described previously.[19] NO_2^--N, NO_3^--N, and NH_4^+-N were determined according to standard methods.[18] Fe(II) concentration was analyzed according to the procedure described by Ding et al.[20] Briefly, 1 mL of the slurry was immediately added to 5 mL of HCl solution (1 mol/L) in a sealed glass tube. The tube was shaken for 24 h and subsequently centrifuged at 9000 rpm for 20 min. The supernatant was then measured for ferric iron concentration according to the standard method.[18]

2.4. Microbial metagenomic Analysis. At the end of the experiment, digestates from duplicate reactors were mixed thoroughly prior to microbial analysis. Samples from all experimental groups were collected for metagenomic analysis. The total DNA of each microbial sample was extracted using the FastDNA SPIN Kit for soil (MP Biochemicals Inc., US). The DNA extract was fragmented to an average size of about 400 bp using a Covaris M220 (Gene Company Limited, China). Paired-end library was constructed using NEXTFLEX Rapid DNA-Seq (Bioo Scientific, US). Paired-end sequencing was performed on the NovaSeq platform (Illumina, USA) at Majorbio Bio-Pharm Technology Co., Ltd. (Shanghai, China). The data were analyzed on the free online platform of Majorbio Cloud Platform (www.majorbio.com).

2.5. Statistical Analyses. One-way analysis of variance was conducted to compare the experimental data using SPSS 22.0 (US), and statistical significance was accepted at $p < 0.05$.

3. RESULTS AND DISCUSSION

3.1. Methane Production and Organic Removal. The CH_4 production efficiency is a key indicator for evaluating the performance of AD. The CH_4 potentials of fish sludge depend highly upon the type of fish, feed components, and feeding pattern. Mirzoyan et al. reported an accumulative CH_4 yield of 77.5 mL/g-VS_{fed} from up-flow anaerobic sludge blanket (UASB) reactors treating striped bass sludge.[21] Gebauer et al. obtained a higher CH_4 yield of 123.6−266.3 mL/g-VS_{fed} from the fish sludge of Atlantic salmon in a continuous stirred tank reactor (CSTR) at 35 °C.[22] As shown in Figure 1a, the accumulative CH_4 yields in all reactors rose rapidly on day 0 and flattened out after day 20, suggesting that the labile substances were quickly consumed within a short period of 20 days. After 28 days of AD, the accumulative CH_4 yield of the control, Fe(III), Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups reached 71.9, 81.9, 62.2, 43.4, and 37.0 mL/g-VS_{fed}, respectively. The CH_4 yields obtained in this work were comparable to those in the literature. The CH_4 yield of the Fe(III) group was 13.9% higher ($p < 0.05$) than that of the control group, indicating the effectiveness of iron sludge in improving the methanogenesis performance. The phenomenon was consistent with the results obtained by Wang et al., who attained an improved CH_4 yield by 12.6−21.0% during AD of waste-activated sludge (WAS) with the addition of 2 to 8 g/L of Fenton sludge.[23] The poorly crystalline Fe(III) hydroxides in iron sludge could enrich IRBs that are capable of metabolizing complex organic substrates through the DIR process, which promoted the hydrolysis of organic particulates into smaller molecular substances to benefit methanogenesis.[23] In contrast to Fe(III) addition only, aeration showed a negative effect on AD of fish sludge. Compared with the control group, a slightly lower level of CH_4 yield by 13.5% was attained in the Fe(III)-AE5 group, and significantly lower levels of CH_4 yield by 39.6 and 48.5% were observed in the Fe(III)-AE10 and Fe(III)-AE15 groups, respectively. The results were expected since a high O_2 level is known to be toxic to methanogens.[24]

The TCOD concentrations before and after the AD experiments are shown in Figure 1b. The initial TCOD concentration of the sludge was 22166.5 mg/L, the remaining concentrations of which after AD were determined to be 13610.2, 12657.1, 11969.9, 11327.1, and 10972.4 mg/L in the control, Fe(III), Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups, respectively. The corresponding TCOD removal efficiencies were calculated to be 38.6% for the control, 42.9% for Fe(III), 46.0% for Fe(III)-AE5, 48.9% for Fe(III)-AE10, and 50.5% for Fe(III)-AE15, respectively. Apparently, the supplementation of iron sludge significantly increased the TCOD removal efficiencies ($p < 0.05$). As discussed above, the DIR process induced by Fe(III) hydroxides might partially explain the greater organic degradation efficiencies in the Fe(III)-supplemented digesters. It was noticeable that in spite of the significantly lower CH_4 yield from the three aerated groups, the TCOD removal efficiencies of the Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups were 19.2, 26.7, and 30.8% greater than the control group, respectively. The organic removal efficiencies were positively correlated with aeration time, which could be supported by the CO_2 production result displayed in Figure S2. As shown, the accumulative CO_2 yield increased significantly with the aeration time, and these might be explained by the following reasons: (1) the injected O_2 could support the growth and metabolisms of facultative organisms, which effectively enhanced the degradation of organics[14]; (2) a certain proportion of the regenerated Fe(III) via O_2 oxidation could engage in the DIR reactions; and (3) the Fe(II)/Fe(III) cycle could drive the Feammox process to generate more NO_x^-, which was further converted into N_2 via heterotrophic denitrification, resulting in the consumption of greater amounts of organics.

Figure 1c demonstrates the variations of the sCOD with time. The sCOD concentrations in different reactors varied similarly, which increased rapidly from the beginning of the experiment and hit the top on day 4, mainly caused by the hydrolysis of macromolecular substances. In the following days, the sCOD in all groups gradually decreased to a low level after day 20, suggesting that most labile substrates were

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

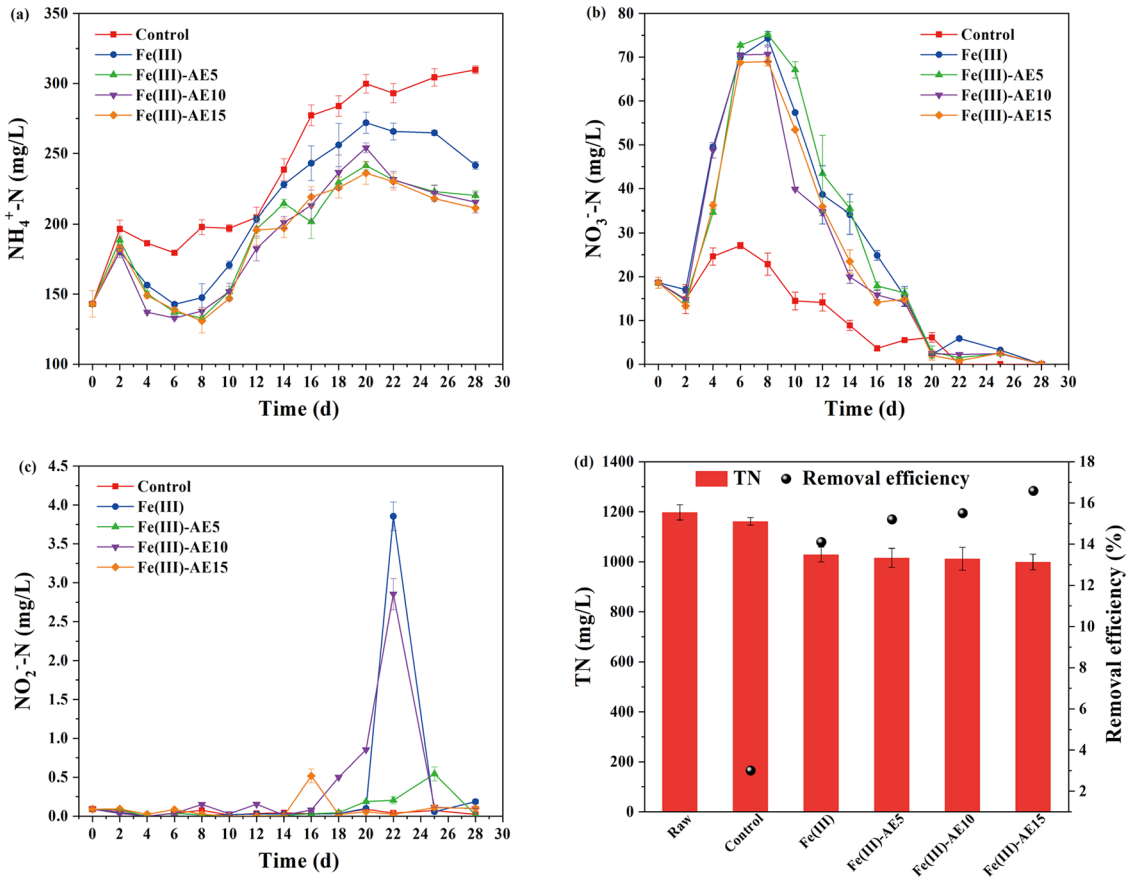


Figure 2. Concentrations of (a) NH_4^+-N, (b) NO_3^--N, (c) NO_2^--N, and (d) TN.

effectively consumed by the microorganisms. These results agreed with the cessation of methanogenesis after day 20. The variations of soluble PS and PN are illustrated in Figure 1d,e, respectively. Both substances exhibited a decreasing trend with time, and their concentrations in the four experimental groups with iron sludge supplementation were significantly lower than those in the control group ($p < 0.05$), which might be explained by the promotion effect of iron sludge on the degradation of soluble PS and PN into smaller molecular organics, which facilitated their subsequent utilization by bacteria (e.g., acetogens, IRBs, heterotrophic denitrifiers, etc.) or methanogens. As illustrated in Figure 1f, acetic acid (HAc) and propionic acid (HPr) were the main components of VFAs, and the trends of VFAs were similar to those of sCOD. The VFAs in all groups increased rapidly within the first 4 days, and then, they gradually decreased until the end of the experiment. Correspondingly, the pH in all groups decreased within the first 4 days and then gradually increased with the consumption of VFAs and stabilized after 16 days (Figure S3). On day 4, the VFA concentration of the Fe(III) group (3061.9 mg/L) was significantly higher than that of the control group (2710.0 mg/L) ($p < 0.05$). This indicated that iron sludge supplementation had facilitated the production of VFAs at the early stage of AD, which could be attributed to the enhanced enzymatic activities involved in hydrolysis and acidification.[25] At the end of AD, the VFA concentration of the Fe(III) group (680.6 mg/L) was lower than that of the control group (901.2 mg/L) ($p < 0.05$), suggesting that iron sludge had accelerated the decomposition of VFAs via methanogenesis or other pathways at the late stage of AD. For the aerated reactors, the highest VFA concentrations were recorded at 2791.6, 2688.0, and 2696.0 mg/L in the Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups on day 4, respectively. These values were quite close to that of the Fe(III) group on the same day. Apparently, aeration did not significantly hinder the acidification performance. At the late stage of AD (days 16−28), the concentrations of VFAs in the aerated reactors were higher than that of the control group, especially the Fe(III)-AE10 and Fe(III)-AE15 groups. This indicated that higher aeration intensities significantly inhibited the conversion of VFAs to CH_4 (Figure 1a).

3.2. Nitrogen Removal. Figure 2a depicts the changes in NH_4^+-N in different groups. During AD of protein-rich fish sludge, protein is decomposed stepwise into polypeptides, amino acids, and short-chain organic acids. Ammonia accumulates gradually with the anaerobic decomposition process. In the control group, NH_4^+-N increased gradually from an initial value of 143.1 to 309.8 mg/L by the end of AD. The addition of iron sludge contributed to a lower level of NH_4^+-N in the Fe(III) group ($p < 0.05$), with a final NH_4^+-N concentration of 241.6 mg/L being recorded at the end of AD, which was 22.0% lower than that of the control. Apparently, the supplementation of Fe(III) led to the loss of NH_4^+-N during AD of the fish sludge. The oxidation of NH_4^+-N in an anaerobic environment is quite challenging due to the lack of appropriate electron acceptors. Anammox process can effectively oxidize NH_4^+-N using NO_2^--N as an electron acceptor under anaerobic conditions, and the existence of Fe(III) may improve the activity of Anammox.[26] However, autotrophic Anammox bacteria are characterized by their slow growth rate and susceptibility to high organic contents.[27]

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

Anammox bacteria will be inhibited at a C/N ratio above 2.0,[28] and they will be seriously suppressed and outcompeted by the heterotrophic bacteria at $C/N > 4.0$.[29] Complete suppression of the Anammox process was observed at TCOD concentrations above 700 mg/L.[30] In this work, the initial C/N ratio (15.7) and TCOD concentration (22,166.5 mg/L) of the mixture were much higher than the inhibitory threshold values of Anammox. Therefore, Anammox reactions were unlikely to occur during AD of fish sludge, and the loss of NH_4^+-N could be attributed to other Fe(III)-inducing ammonia oxidation processes like Feammox. Compared with the case of iron sludge addition only, coupling iron sludge addition with intermittent aeration further decreased the NH_4^+-N concentrations in the Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups. This result was expected since continuous Feammox reactions could be driven by the Fe(II)/Fe(III) cycle with O_2 as the terminal electron acceptor.[15] It should be mentioned that aeration might also induce ammonia stripping and nitrification to accelerate NH_4^+ loss. To quantify the possible contributions of stripping and nitrification, an additional set of AD experiments (AE10 group) were performed, and the variations of NH_4^+-N are displayed in Figure S4. As indicated, the final NH_4^+-N concentration in the AE10 group was 308.7 mg/L, which was very close to that of the control. The result suggested that neither stripping nor nitrification played an important role in NH_4^+-N removal. The greater NH_4^+-N loss with intermittent aeration was mostly attributable to the Feammox process driven by the regenerated Fe(III), and this will be discussed later in Section 3.3. Despite the positive effect of aeration on promoting the Feammox process, an increased aeration time from 5 to 15 min did not significantly lower the system NH_4^+-N concentration ($p > 0.05$). On day 28, the NH_4^+-N concentrations were 220.1, 215.2, and 211.0 mg/L in the Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups, respectively. Feammox is an anoxic process, and a low level of DO favors the reactions.[11] Yao et al. observed a quick decline in the Feammox rate at a DO level above 1.0 mg/L.[31] To assess the possible toxicity of DO to the Feammox bacteria in the intermittent aerated reactors, the variations of DO during one aeration cycle on day 4 were tested (Figure S5). After 5, 10, and 15 min of aeration, the DO concentration increased to 0.06, 0.14, and 0.20 mg/L, respectively. When the aeration was stopped, the DO was quickly consumed by the microorganisms and became undetectable within 30−60 min. These DO levels were not expected to significantly suppress the Feammox process, and the trade-off between oxygen inhibition and Fe(III) regeneration was clearly not the reason for the insignificant improvement in NH_4^+-N removal efficiency at higher aeration intensities. From the experimental results in this work, it was hypothesized that although a sufficient amount of Fe(III) was a prerequisite for the Feammox process, the removal of NH_4^+-N was restricted by the slow Feammox reaction rate. In a review by Xia et al., the N removal rates of nitrification-denitrification, Anammox, and partial denitrification-Anammox were calculated to be 21−58, 106−876, and 310−730 N/m^3/day, respectively.[11] Compared with these N removal processes, the reaction rate of Feammox is much slower. Yang et al. reported that it took 90 days to achieve 98.5% of TN removal from a Feammox reactor (initial TN of 1021.5 mg/L),[15] and the N removal rate was calculated to be 11.2 g N/m^3/day. In this study, the N removal rates in the Fe(III)-supplemented reactors during days 2−6 were calculated to be 9.2−12.8 g N/m^3/day. To further improve the Feammox performance, future research efforts should focus on the acceleration of the reactions by optimizing the operational parameters or establishing direct interspecies electron transfer (DIET) by adding electron shuttles.

It is worth noting that the NH_4^+-N concentrations in the four Fe(III)-supplemented digesters showed a trend of rapid decline since day 20. As discussed above in Section 3.1, a certain proportion of Fe(III) could engage in the DIR process. Feammox was not dominant in Fe(III) reduction at high organic concentrations because it was at a disadvantage when competing with DIR for electron acceptors.[32] Ding et al. reported that only 2.2% of Fe(III) reduction was directly attributed to the Feammox process in paddy soils.[20] Therefore, Fe(III) in digesters treating fish sludge preferentially oxidized organics rather than ammonia. By day 20, most easily biodegradable organics were consumed, which was evidenced by the flattening of accumulative CH_4 yield and sCOD (Figure 1a,c), and this could explain the rapid decline of NH_4^+-N after day 20 when Feammox started to gain advantages.

Figure 2b illustrates the changes in NO_3^--N. The initial NO_3^--N concentration in all groups was 18.6 mg/L. In the control group, NO_3^--N increased to a maximum value of 27.1 mg/L by day 6, followed by a gradual decline until the end of AD. The slight increase of NO_3^--N during days 0−6 in the control group might be attributed to Feammox process induced by the small amount of Fe(III) in the inoculum and raw fish sludge. In comparison, NO_3^--N concentrations of the four Fe(III)-supplemented reactors increased from 13.3 to 17.0 to 69.0−74.3 mg/L during days 2−8, which was in accordance with the decrease of NH_4^+-N in these reactors from 180.4 to 188.6 to 130.9−147.5 mg/L during the same period (Figure 2a). Mass balance analysis indicated that NO_3^--N was the main product of Feammox in the Fe(III)-supplemented reactors. After day 8, the NO_3^--N concentrations in the four reactors dropped quickly to below 3.0 mg/L by day 20 and leveled off after that, confirming the occurrence of some biological or biochemical NO_3^--N removal processes like denitrification or NO_x^--dependent Fe(II) oxidation (NDFO). Theoretically, the generated NO_x^- can be converted into N_2 via the NDFO process, but the autotrophic NDFO bacteria did not have a growth advantage in AD reactors with high organic contents.[11] Therefore, it was reasonable to deduce that denitrification was the main pathway for the NO_3^--N removal pathway. Most of the time, NO_2^--N in all reactors was maintained at a low level of nearly undetectable though the Fe(III) and Fe(III)-AE10 groups attained relatively high concentrations during days 20−25, with a maximum value of 3.9 and 2.9 mg/L being recorded on day 22, respectively (Figure 2c).

The initial and final TN contents were measured to further investigate the effects of iron sludge addition and intermittent aeration on N removal (Figure 2d). The initial TN concentration was 1197.7 mg/L in all groups. After digestion, it decreased to 1161.7, 1028.8, 1015.7, 1012.1, and 999.1 mg/L in the control, Fe(III), Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups, corresponding to a removal efficiency of 3.0, 14.1, 15.2, 15.5, and 16.5%, respectively. The decreased TN by 3% in the control group might be attributed to the small amount of Fe(III) in the inoculum derived from the incubation process or the original fish feed. Fe(III) supplementation and intermittent aeration greatly promoted the removal of N during AD.

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

3.3. Changes of Iron. Only a proportion of the iron content in solid iron sludge could be utilized by the IRBs, and the supplemented iron sludge mixed in the fish sludge was difficult to separate and tested. Therefore, the variations of Fe(II) were monitored in this work instead of the total iron, as the total iron concentrations could not reflect the changes of iron induced by intermittent aeration and Feammox/DIR reactions.[33] It was reported that in the aerated solutions, the half-life of dissolved Fe(II) was less than 5 min at circumneutral pH.[34] In this study, the system pH fluctuated between 6.7 to 7.6 throughout the experiment in all reactors (Figure S3). As shown in Figure 3, the concentration of Fe(II)

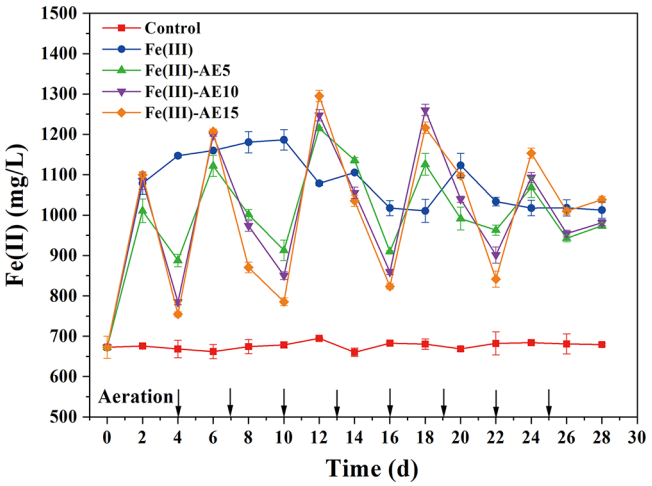


Figure 3. Fe(II) concentrations in different groups during the AD process.

in the control group remained constant at about 670 ± 10 mg/L throughout the experiment. A slight fluctuation of Fe(II) between 1010.4 and 1186.5 mg/L during days 2−28 was observed in the Fe(III) group. In sharp contrast, intermittent aeration contributed to a much greater extent of Fe(II) fluctuation. For example, the Fe(II) concentrations in the three aerated reactors increased to about 1100 mg/L on day 2 due to Feammox and/or DIR reactions; then, it decreased to 887.8, 784.3, and 754.5 mg/L on day 4 after being aerated for 5, 10, and 15 min, respectively. Rapid Fe(II) oxidation occurred in the reactors undergoing redox cycling, leading to the regeneration of Fe(III). The variation of Fe(II) confirmed the occurrence of the Fe(II)/Fe(III) cycle induced by intermittent aeration. Remarkably, the Fe(II) levels were significantly different among the three reactors with different aeration intensities ($p < 0.05$). The ranges of Fe(II) fluctuation followed an order of Fe(III)-AE15 > Fe(III)-AE10 > Fe(III)-AE5, confirming that a higher aeration intensity indeed promoted the oxidation of Fe(II) to Fe(III). It was noticeable that the amplitude of Fe(II) fluctuation became weaker with time. On the one hand, the active (amorphous) Fe(III) (hydr)oxide could be transformed into thermodynamically more stable crystalline minerals (e.g., magnetite and Fe_2O_3) with time,[15] resulting in the decrease of microbial reducible Fe(III). On the other hand, the Fe(II) could attach to the surface of the Fe(III) compounds and be converted to iron oxyhydroxide Fe(III), providing the nuclei for the formation of secondary minerals and facilitating the uptake of aqueous Fe(II), which further decreased the Fe(II) during redox cycling.[12]

3.4. Microbial Metagenomic Analysis. To further reveal the effects of iron sludge addition and its combination with intermittent aeration on N removal, we annotated functional genes of the microbial samples according to the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. Specifically, functional genes involved in five N metabolism processes were investigated, namely denitrification, dissimilatory nitrate reduction, assimilatory nitrate reduction, N fixation, and nitrification processes (Figure 4a). The abundances of specific genes are indicated in Figure 4b. As shown, the abundances of genes relative to nitrification were very low, and anammox-related genes were not detected at all, suggesting that neither nitrification nor Anammox was the main pathway for ammonia conversion. In contrast, the total abundances of denitrification-related genes were the highest among the five N conversion processes studied, confirming that denitrification played an important role in N removal from the reactors. The abundances of denitrification-related genes were generally higher in the Fe(III), Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups with Fe(III) supplementation. These results suggested that the supplementation of Fe(III) must have induced certain reactions to convert NH_4^+ into NO_x^- and subsequently drive the denitrification process. It is worth mentioning that functional genes and proteins associated with the Feammox process have not been identified yet;[11] therefore, the current study could not directly prove the existence of Feammox-related genes. Despite this, the above results could provide circumstantial evidence to support the assumption that Fe(III)-induced Feammox coupled to denitrification was the main N removal pathway during AD of fish sludge. The abundances of genes relative to assimilatory nitrate reduction (*narB*, *nirA*, and *nasB*) and N fixation (*nif D/H/K*) were significantly lower compared with the genes involved in dissimilatory nitrate reduction, indicating that dissimilatory nitrate reduction was mainly responsible for the regeneration of NH_4^+-N in this study. As seen, the supplementation of Fe(III) significantly decreased the abundance of *nirB*, a key gene encoding NO_2^- reductase during dissimilatory nitrate reduction, which was beneficial to the overall N removal performance.

To study the roles of different bacteria in N metabolism, we further examined the sources of related functional genes. Figure 4c,d displays the contributions of major taxonomic groups to denitrification and dissimilatory nitrate reduction functions at the phylum and the genus levels, respectively. As shown, the denitrification and dissimilatory nitrate reduction functions were mostly contributed by the same groups of bacteria. The top four phyla contributing to denitrification function in all reactors were *Proteobacteria* (34.2−40.5%), *Chloroflexi* (23.9−28.1%), *Bacteroidetes* (10.6−13.4%), and *Planctomycetes* (7.0−9.3%). Similarly, the above four phyla contributed to 28.1−33.6, 27.7−33.2, 13.6−15.8, and 6.9−8.9% of the overall dissimilatory nitrate reduction function, respectively. On the genus level, *unclassified_p__Chloroflexi*, *unclassified_o__Bacteroidales*, *unclassified_f__Anaerolineaceae*, and *unclassified_p__Betaproteobacteria* were the main contributors to the denitrification and dissimilatory nitrate reduction functions.

The microbial communities in different experimental groups were studied at the phylum level (Figure 5a) and genus level (Figure 5b), respectively. As shown in Figure 5a, *Chloroflexi* was the predominant phylum, accounting for 29.8−32.9% of the overall microbial abundances, followed by *Proteobacteria*

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

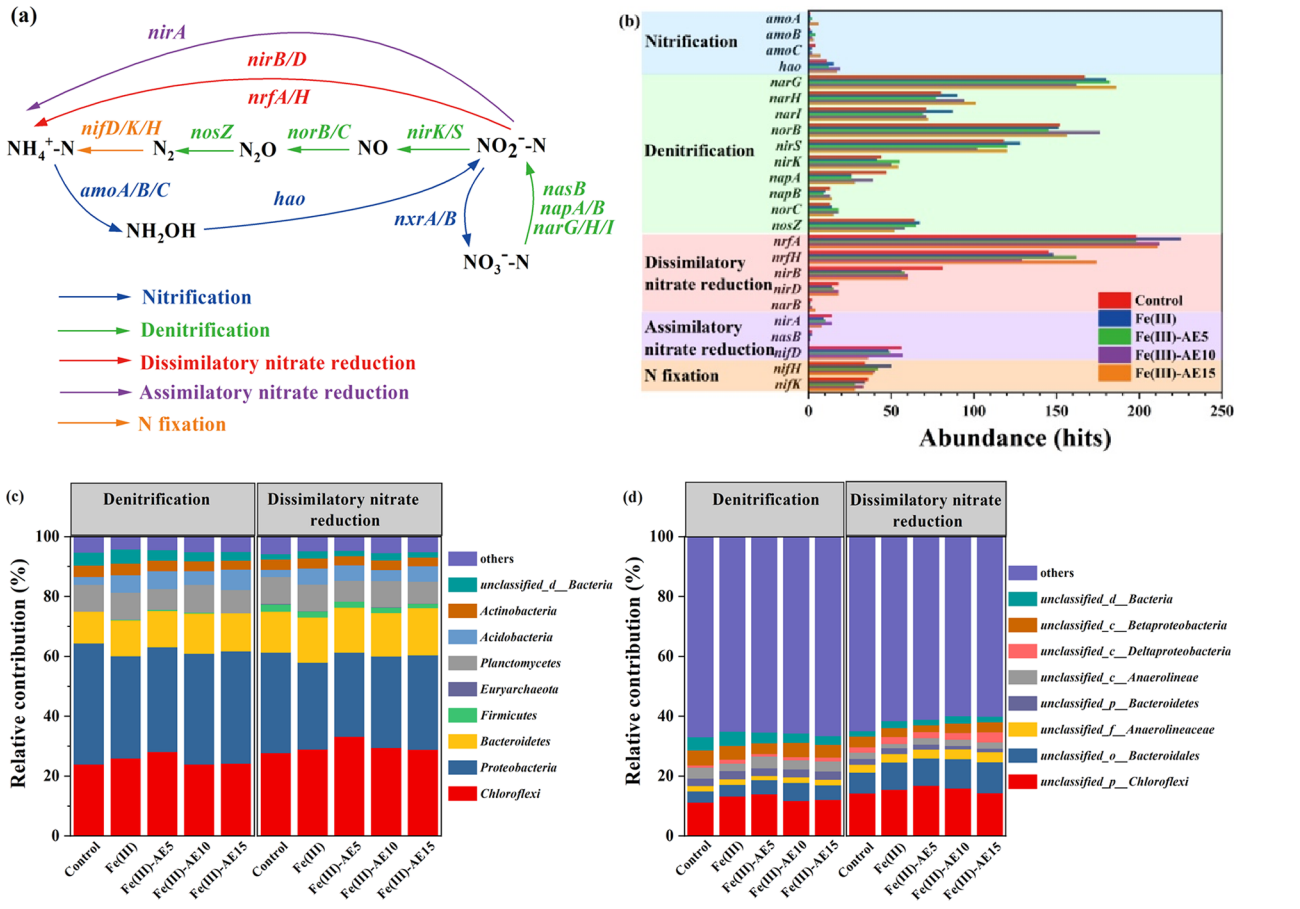


Figure 4. (a) N conversion pathways and related functional genes; (b) abundances of specific functional genes related to N metabolism; contributions of major taxonomic groups to key N conversion functions (c) at the phylum level and (d) the genus level, respectively.

(16.5−19.0%) and *Bacteroidetes* (14.0−16.7%). As discussed above, these three phyla were the major contributors to the N removal function in this work. They are also well-known hydrolytic-fermentative bacteria in AD systems.[35] Zhou et al. reported that these three phyla were the main phyla in an up-flow sludge bed biofilm reactor and were responsible for organic degradation and N transformation.[36] Overall, there was no significant difference in the phylum-level microbial communities among the five groups.

The distributions of the genera are illustrated in Figure 5b. Typical nitrifying bacteria such as *Nitrosomonas*, *Nitrosococcus*, and *Nitrospira*,[37] ammonia-oxidizing archaea such as *Nitrosopumilus maritimus* and *Nitrososphaera viennensis*,[38] and Anammox bacteria including *Kuenenia*, *Brocadia*, *Anammoxoglobus*, *Jettenia*, and *Scalindua*[39] were not detected from any of the reactors. Nitrifying bacteria usually grow at a DO level exceeding 0.4 mg/L,[6] ammonia-oxidizing archaea prefer an acidic environment,[40] and Anammox bacteria can be seriously suppressed at high organic contents.[30] In this study, the DO concentrations in all reactors were not higher than 0.2 mg/L, the pH levels were maintained at 6.7−7.6, and the TCOD levels were 15−80 times higher than the inhibitory threshold for Anammox bacteria, which were not suitable for the growth of nitrifying bacteria, ammonia-oxidizing archaea, or Anammox bacteria. Currently, the investigation on Feammox-related microbes is still in the infancy, but studies have suggested that IRBs (e.g., *Longilinea*, *Bacteroides*, *Deltaproteobacteria*, *Acidobacteria*, and *Anaerolinea*) are the primary functional bacteria mediating Feammox.[11,17,41] The relative abundances of IRBs are shown in Figure 5c. As shown, the total abundance of IRBs increased by 5.0−9.4% in the four Fe(III)-supplemented reactors when compared with the control group, confirming that the addition of iron sludge could enrich the IRBs, and this was in agreement with the increased organic and NH_4^+-N removal efficiencies. Only one type of methanogen (*Methanothrix*) was detected and identified from all digesters. Compared with the control, the relative abundance of *Methanothrix* was increased by 3.3% in the Fe(III) reactor, while it decreased by 0.2, 2.6, and 13.8% in the Fe(III)-AE5, Fe(III)-AE10, and Fe(III)-AE15 groups, respectively (Figure 5d). The abundances of *Methanothrix* in different reactors were in line with the CH_4 production performances.

3.5. Prospects. Iron sludge is a kind of solid waste produced from the treatment of wastewater, the in situ utilization of which is beneficial. This study demonstrated the feasibility of SMFD in a single reactor treating fish sludge by iron sludge addition and periodic air supply. Compared with the conventional treatment processes (AD followed by biological N removal), the SMFD system has the advantages of a smaller footprint and much lower construction and operational costs. The energy consumption of the complete nitrification-denitrification process is about 4.6 kWh/kg N.[11] While the energy consumption for aeration in the SMFD system is negligible. The volumetric lower heating value of CH_4 is 35.8 MJ/m^3.[42] About 30% of the heat released from CH_4 combustion can be converted into electricity, and the electricity production from CH_4 is calculated to be 3.0 kWh/m^3. Although intermittent aeration showed a negative impact

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

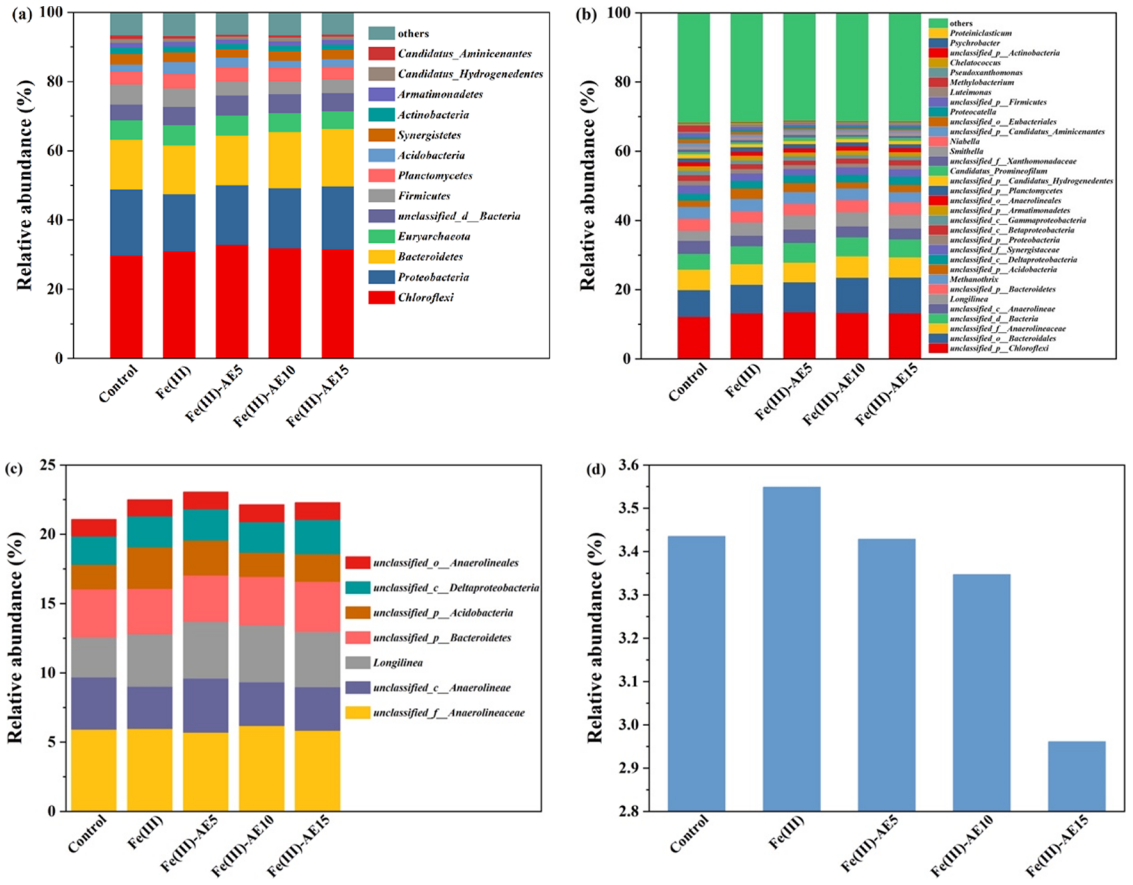


Figure 5. Microbial community distributions at (a) phylum level and (b) genus level, respectively; relative abundances of (c) IRBs and (d) *Methanothrix*.

on the CH_4 yield, this was acceptable compared with the energy savings for N removal. Under the optimized aeration conditions (5 min every 3 days), the electricity saved for N removal was 44.7% higher than the decreased amount recovered from CH_4 production. Despite the merits of the integrated system, the overall N removal efficiency was greatly limited by the slow electron transfer between NH_4^+-N and Fe(III). To improve the overall N removal performance, future research efforts should focus on further optimization of process parameters (e.g., pH, temperature, C/N ratio, etc.) for enhancing the activities of Feammox bacteria and acceleration of interspecies electron transfer by adding redox-active compounds or chelating agents.

Anaerobic decomposition of N-rich substrates such as fish sludge, WAS, and animal manure often encounters elevated concentrations of ammonia. Ammonia inhibition is one of the most common causes of instability in industrial biogas plants. The SMFD system provides a cost-effective alternative to alleviate ammonia inhibition during AD. It is noteworthy that Fe(III) is preferentially utilized in the DIR process over ammonia oxidation. In this study, with the depletion of labile organics, the removal of NH_4^+-N became faster at the late stage of methanogenesis, and a long retention time was required for sufficient ammonia removal from the AD reactors treating fish sludge. This phenomenon suggested that the SMFD system might not be the most suitable for the treatment of substrates rich in easily biodegradable organics. However, it has great potential for the treatment of low C/N wastewater or refractory organic wastewater with high NH_4^+-N contents such as digestates or pharmaceutical wastewater. The DIR process has been proven to be effective for the anaerobic degradation of refractory organic compounds. In addition, intermittent aeration can facilitate the growth of facultative bacteria, thereby promoting the hydrolysis of macromolecular compounds. By coupling iron sludge addition with intermittent aeration, the N in wastewater can be removed in situ through Feammox and denitrification, and the organic C can be further degraded and recovered as CH_4.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.3c03009.

Main characteristics of the fish sludge, inoculum, and mixture; schematic of the reactors used in this study; accumulative CO_2 yield in different reactors; variations of pH in different groups during AD; changes of NH_4^+-N in the control and AE10 groups; and changes of DO during one aeration cycle on day 4 (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Weiwei Huang − *Key Laboratory of Agro-Forestry Environmental Processes and Ecological Regulation of Hainan Province, College of Ecology and Environment, Hainan University, Haikou 570228, China*; Email: huang05106114@163.com

https://doi.org/10.1021/acs.est.3c03009
Environ. Sci. Technol. 2023, 57, 15065−15075

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

Xuesong Yi − *Key Laboratory of Agro-Forestry Environmental Processes and Ecological Regulation of Hainan Province, College of Ecology and Environment, Hainan University, Haikou 570228, China;* orcid.org/0000-0003-4797-9562; Email: cedar401@163.com

Authors

Qing Xia − *Key Laboratory of Agro-Forestry Environmental Processes and Ecological Regulation of Hainan Province, College of Ecology and Environment, Hainan University, Haikou 570228, China;* orcid.org/0000-0003-4788-3683

Fei Liu − *Key Laboratory of Agro-Forestry Environmental Processes and Ecological Regulation of Hainan Province, College of Ecology and Environment, Hainan University, Haikou 570228, China*

Shengrui Sun − *Key Laboratory of Agro-Forestry Environmental Processes and Ecological Regulation of Hainan Province, College of Ecology and Environment, Hainan University, Haikou 570228, China*

Wenli Huang − *MOE Key Laboratory of Pollution Process and Environmental Criteria, College of Environmental Science and Engineering, Nankai University, Tianjin 300071, China*

Ziwen Zhao − *Ministry of Ecology and Environment, South China Institute of Environmental Sciences, Guangzhou 510345, China*

Fei Yang − *Key Laboratory of Agro-Forestry Environmental Processes and Ecological Regulation of Hainan Province, College of Ecology and Environment, Hainan University, Haikou 570228, China*

Zhongfang Lei − *Graduate School of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Ibaraki 305-8572, Japan;* orcid.org/0000-0002-0479-6179

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.est.3c03009

Author Contributions

$^{\perp}$Q.X. and F.L. contributed equally to the paper.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the National Natural Science Foundation of China (52060005).

■ REFERENCES

(1) Hu, Y.; Liu, S.; Wang, X.; Zhang, S.; Hu, T.; Wang, X.; Wang, C.; Wu, J.; Xu, L.; Xu, G.; Hu, F. Enhanced anaerobic digestion of kitchen waste at different solids content by alkali pretreatment and bentonite addition: Methane production enhancement and microbial mechanism. *Bioresour. Technol.* **2023**, *369*, No. 128369.

(2) He, L.; Lin, Z.; Zhu, K.; Wang, Y.; He, X.; Zhou, J. Mesophilic condition favors simultaneous partial nitrification and denitrification (SPND) and anammox for carbon and nitrogen removal from anaerobic digestate food waste effluent. *Sci. Total Environ.* **2022**, *816*, No. 151498.

(3) Qian, Y.; Shen, J.; Chen, F.; Guo, Y.; Qin, Y.; Li, Y. Y. Increasing nitrogen and organic matter removal from swine manure digestate by including pre-denitrification and recirculation in single-stage partial nitritation/anammox. *Bioresour. Technol.* **2023**, *367*, No. 128229.

(4) Qiu, J.; Lü, F.; Zhang, H.; Huang, Y.; Shao, L.; He, P. Persistence of native and bio-derived molecules of dissolved organic matters during simultaneous denitrification and methanogenesis for fresh waste leachate. *Water Res.* **2020**, *175*, No. 115705.

(5) Wu, X.; Wang, C.; Wang, D.; Huang, Y. X.; Yuan, S.; Meng, F. Simultaneous methanogenesis and denitrification coupled with nitrifying biofilm for high-strength wastewater treatment: Performance and microbial mechanisms. *Water Res.* **2022**, *225*, No. 119163.

(6) Guo, J. H.; Peng, Y. Z.; Wang, S. Y.; Zheng, Y. N.; Huang, H. J.; Wang, Z. W. Long-term effect of dissolved oxygen on partial nitrification performance and microbial community structure. *Bioresour. Technol.* **2009**, *100*, 2796−2802.

(7) Ding, B.; Zhang, H.; Luo, W.; Sun, S.; Cheng, F.; Li, Z. Nitrogen loss through denitrification, anammox and Feammox in a paddy soil. *Sci. Total Environ.* **2021**, *773*, No. 145601.

(8) Ding, B. J.; Li, Z. K.; Qin, Y. B. Nitrogen loss from anaerobic ammonium oxidation coupled to Iron(III) reduction in a riparian zone. *Environ. Pollut.* **2017**, *231*, 379−386.

(9) Yi, B.; Wang, H. H.; Zhang, Q. C.; Jin, H.; Abbas, T.; Li, Y.; Liu, Y. M.; Di, H. J. Alteration of gaseous nitrogen losses via anaerobic ammonium oxidation coupled with ferric reduction from paddy soils in Southern China. *Sci. Total Environ.* **2019**, *652*, 1139−1147.

(10) Zhu, T. t.; Lai, W. x.; Zhang, Y. B.; Liu, Y. w. Feammox process driven anaerobic ammonium removal of wastewater treatment under supplementing Fe(III) compounds. *Sci. Total Environ.* **2022**, *804*, No. 149965.

(11) Xia, Q.; Ai, Z.; Huang, W.; Yang, F.; Liu, F.; Lei, Z.; Huang, W. Recent progress in applications of Feammox technology for nitrogen removal from wastewaters: A review. *Bioresour. Technol.* **2022**, *362*, No. 127868.

(12) Boland, D. D.; Collins, R. N.; Miller, C. J.; Glover, C. J.; Waite, T. D. Effect of Solution and Solid-Phase Conditions on the Fe(II)-Accelerated Transformation of Ferrihydrite to Lepidocrocite and goethite. *Environ. Sci. Technol.* **2014**, *48*, 5477−5485.

(13) Sugimori, H.; Kanzaki, Y.; Yokota, K.; Murakami, T. Nonlinear dependence of the oxidation rate of Fe(II) on dissolved oxygen under low-O_2 conditions in aqueous solutions. *J. Mineral. Petrol. Sci.* **2011**, *106*, 142−152.

(14) Chen, Q.; Wu, W.; Qi, D.; Ding, Y.; Zhao, Z. Review on microaeration-based anaerobic digestion: State of the art, challenges, and prospectives. *Sci. Total Environ.* **2020**, *710*, No. 136388.

(15) Yang, Y. F.; Zhao, Z. Q.; Zhang, Y. B. Anaerobic Ammonium Removal Pathway Driven by the Fe(II)/Fe(III) Cycle through Intermittent Aeration. *Environ. Sci. Technol.* **2021**, *55*, 7615−7623.

(16) Ma, D.; Wang, J.; Li, H.; Che, J.; Yue, Z. Simultaneous removal of COD and NH_4^+-N from domestic sewage by a single-stage up-flow anaerobic biological filter based on Feammox. *Environ. Pollut.* **2022**, *314*, No. 120213.

(17) Sun, Y.; Wang, M.; Liang, L.; Sun, C.; Wang, X.; Wang, Z.; Zhang, Y. Continuously feeding fenton sludge into anaerobic digesters: Iron species change and operating stability. *Water Res.* **2022**, *226*, No. 119283.

(18) American Public Health Association (APHA), American Water Works Association (AWWA) and Water Environment Federation (WEF). *Standard Methods for the Examination of Water and Wastewater*, 23rd ed.; Washington, D.C., USA. 2017.

(19) Huang, W. W.; Yang, F.; Huang, W. L.; Wang, D. X.; Lei, Z. F.; Zhang, Z. Y. Weak magnetic field significantly enhances methane production from a digester supplemented with zero valent iron. *Bioresour. Technol.* **2019**, *282*, 202−210.

(20) Ding, L. J.; An, X. L.; Li, S.; Zhang, G. L.; Zhu, Y. G. Nitrogen Loss through Anaerobic Ammonium Oxidation Coupled to Iron Reduction from Paddy Soils in a Chronosequence. *Environ. Sci. Technol.* **2014**, *48*, 10641−10647.

(21) Mirzoyan, N.; Parnes, S.; Singer, A.; Tal, Y.; Sowers, K.; Gross, A. Quality of brackish aquaculture sludge and its suitability for anaerobic digestion and methane production in an upflow anaerobic sludge blanket (UASB) reactor. *Aquaculture* **2008**, *279*, 35−41.

(22) Gebauer, R. Mesophilic anaerobic treatment of sludge from saline fish farm effluents with biogas production. *Bioresour. Technol.* **2004**, *93*, 155−167.

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026

(23) Wang, M. W.; Zhao, Z. Q.; Zhang, Y. B. Sustainable Strategy for Enhancing Anaerobic Digestion of Waste Activated Sludge: Driving dissimilatory Iron Reduction with Fenton Sludge. *ACS Sustainable Chem. Eng.* **2018**, *6*, 2220−2230.
(24) Yang, H.; Deng, L.; Wu, J.; Wang, W.; Zheng, D.; Wang, Z.; Liu, Y. Intermittent air mixing system for anaerobic digestion of animal wastewater: Operating conditions and full-scale validation. *Bioresour. Technol.* **2021**, *335*, No. 125304.
(25) Liang, J.; Luo, L.; Li, D.; Varjani, S.; Xu, Y.; Wong, J. W. C. Promoting anaerobic co-digestion of sewage sludge and food waste with different types of conductive materials: Performance, stability, and underlying mechanism. *Bioresour. Technol.* **2021**, *337*, No. 125384.
(26) Wang, H.; Fan, Y.; Zhou, M.; Wang, W.; Li, X.; Wang, Y. Function of Fe(III)-minerals in the enhancement of anammox performance exploiting integrated network and metagenomics analyses. *Water Res.* **2022**, *210*, No. 117998.
(27) Zhang, Q.; Lin, J. G.; Kong, Z.; Zhang, Y. A critical review of exogenous additives for improving the anammox process. *Sci. Total Environ.* **2022**, *833*, No. 155074.
(28) Singh, V.; Ormeci, B.; Mishra, S.; Hussain, A. Simultaneous partial Nitrification, ANAMMOX and denitrification (SNAD) − A review of critical operating parameters and reactor configurations. *Chem. Eng. J.* **2022**, *433*, No. 133677.
(29) Ni, S. Q.; Ni, J. Y.; Hu, D. L.; Sung, S. W. Effect of organic matter on the performance of granular anammox process. *Bioresour. Technol.* **2012**, *110*, 701−705.
(30) Arora, A. S.; Nawaz, A.; Qyyum, M. A.; Ismail, S.; Aslam, M.; Tawfik, A.; Yun, C. M.; Lee, M. Energy saving anammox technology-based nitrogen removal and bioenergy recovery from wastewater: Inhibition mechanisms, state-of-the-art control strategies, and prospects. *Renew. Sustain. Energy Rev.* **2021**, *135*, No. 110126.
(31) Yao, Z. B.; Wang, C. H.; Song, N.; Wang, C. L.; Jiang, H. L. Oxidation of ammonium in aerobic wastewater by anoxic ferric iron-dependent ammonium oxidation (Feammox) in a biofilm reactor. *Desalin. Water Treat.* **2020**, *173*, 197−206.
(32) Li, X. F.; Hou, L. J.; Liu, M.; Zheng, Y. L.; Yin, G. Y.; Lin, X. B.; Cheng, L.; Li, Y.; Hu, X. T. Evidence of Nitrogen Loss from Anaerobic Ammonium Oxidation Coupled with Ferric Iron Reduction in an Intertidal Wetland. *Environ. Sci. Technol.* **2015**, *49*, 11560−11568.
(33) Yang, Y. F.; Zhang, Y. B.; Li, Y.; Zhao, H. M.; Peng, H. Nitrogen removal during anaerobic digestion of wasted activated sludge under supplementing Fe(III) compounds. *Chem. Eng. J.* **2018**, *332*, 711−716.
(34) Mejia, J.; Roden, E. E.; Ginder-Vogel, M. Influence of Oxygen and Nitrate on Fe (Hydr)Oxide Mineral Transformation and Soil Microbial Communities during Redox Cycling. *Environ. Sci. Technol.* **2016**, *50*, 3580−3588.
(35) Wang, C. Q.; Liu, Y.; Jin, S.; Chen, H.; Xu, X. Y.; Wang, Z. R.; Xing, B.; Zhu, L. Responsiveness extracellular electron transfer (EET) enhancement of anaerobic digestion system during start-up and starvation recovery stages via magnetite addition. *Bioresour. Technol.* **2019**, *272*, 162−170.
(36) Zhou, X.; Yin, Z. Y.; Ge, D. L.; Wang, G. L.; Zhang, W. Metagenome metabolic analysis revealing the mechanism of simultaneous methanogenesis, aerobic methane oxidation and denitrification (SMAMOD) in a microaerobic up-flow sludge bed biofilm reactor. *J. Chem. Technol. Biol.* **2020**, *95*, 2229−2236, DOI: 10.1002/jctb.6410.
(37) Thakur, I. S.; Medhi, K. Nitrification and denitrification processes for mitigation of nitrous oxide from waste water treatment plants for biovalorization: Challenges and opportunities. *Bioresour. Technol.* **2019**, *282*, 502−513.
(38) You, J.; Das, A.; Dolan, E. M.; Hu, Z. Q. Ammonia-oxidizing archaea involved in nitrogen removal. *Water Res.* **2009**, *43*, 1801−1809.
(39) Wu, P.; Chen, J.; Garlapati, V. K.; Zhang, X.; Wani Victor Jenario, F.; Li, X.; Liu, W.; Chen, C.; Aminabhavi, T. M.; Zhang, X. Novel insights into Anammox-based processes: A critical review. *Chem. Eng. J.* **2022**, *444*, No. 136534.
(40) Huang, S.; Chen, C.; Peng, X. C.; Jaffé, P. R. Environmental factors affecting the presence of *Acidimicrobiaceae* and ammonium removal under iron-reducing conditions in soil environments. *Soil Biol. Biochem.* **2016**, *98*, 148−158.
(41) Pang, S.; Li, N.; Luo, H.; Luo, X.; Shen, T.; Yang, Y.; Jiang, J. Autotrophic Fe-Driven Biological Nitrogen Removal Technologies for Sustainable Wastewater Treatment. *Front. Microbiol.* **2022**, *13*, No. 895409.
(42) Mertins, A.; Heiker, M.; Rosenberger, S.; Wawer, T. Competition in the conversion of the gas grid: Is the future of biogas biomethane or hydrogen? *Int. J. Hydrogen Energy* **2023**, *48*, 32469.

Downloaded from pubs.acs.org/esthag/article-pdf/57/40/15065/2647978/es3c03009.pdf by HONG KONG UNIV SCIENCE TECHLGY user on 20 October 2026